



Sir Syed CASE
Institute of Technology

Assignment 2

Artificial Intelligence

SUBMITTED BY:

UBAID AHMAD

ROLL NO:

2410-0011

SUBMITTED TO:

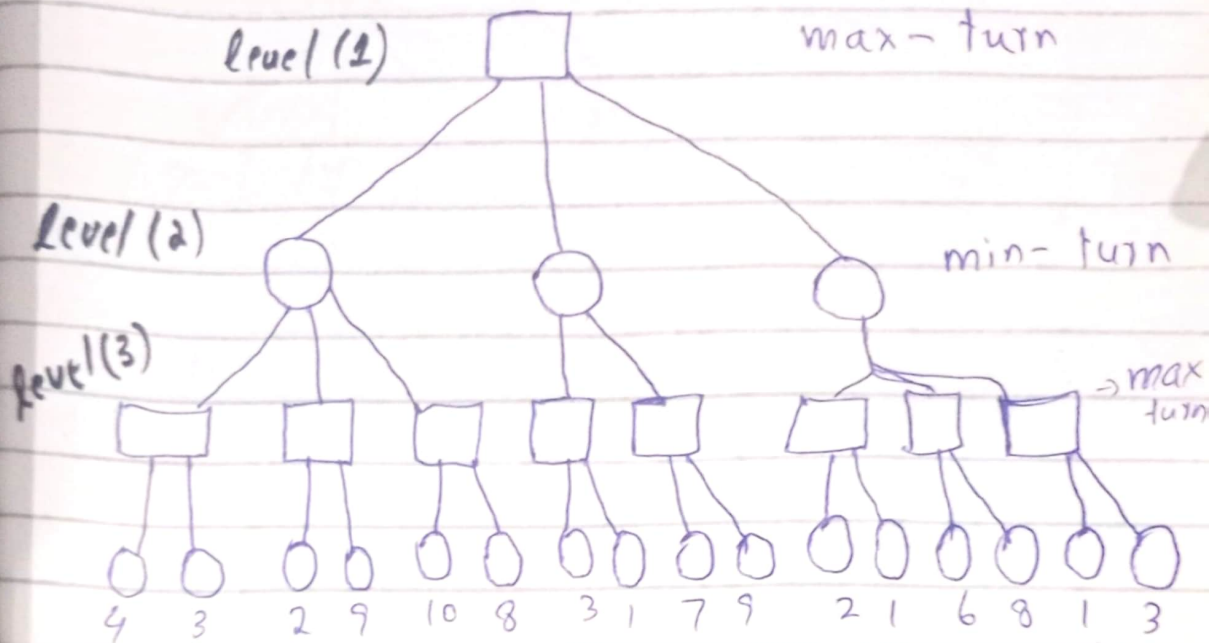
Sir Ashfaq Ahmad

DATE:

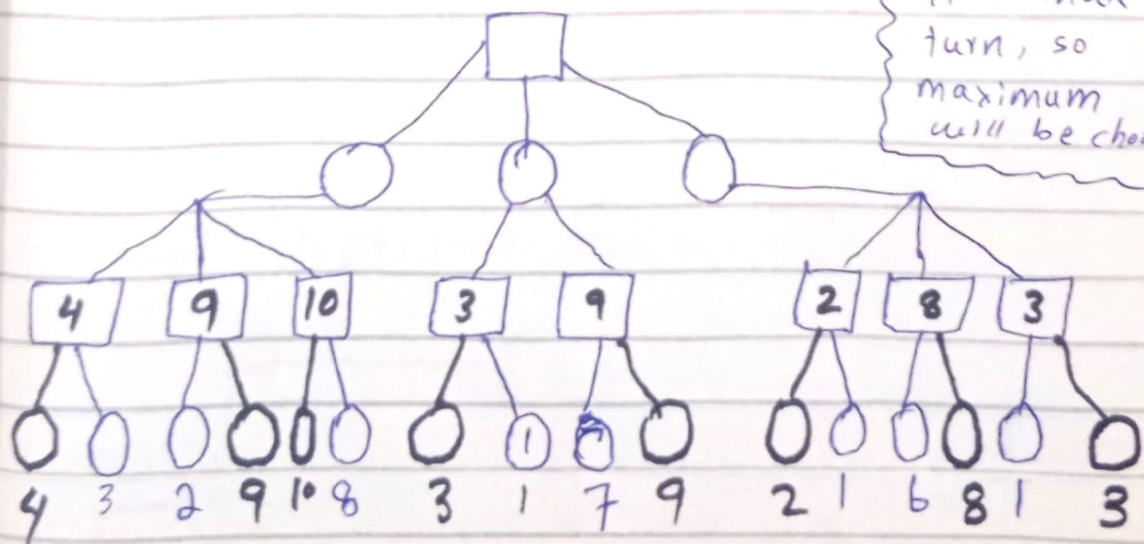
27/04/2026

(QUESTION 1 MINMAX)

Initial Tree:

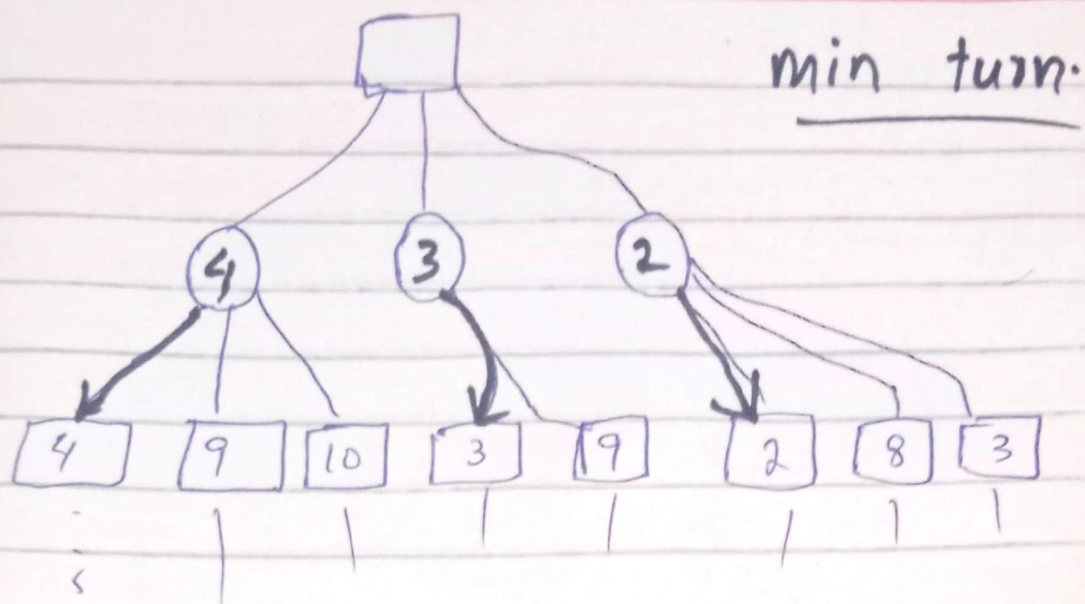


Step: 1 starting from leaf nodes
level 3



Step (2) Level: 2 It is min turn.

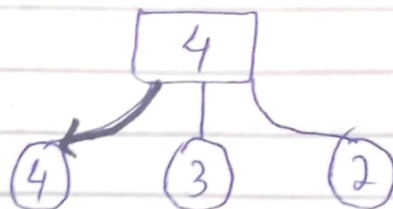
- min has 8 options, 3 moves.
- min will choose least (minimum)
- so it chooses 4, 3, 2.



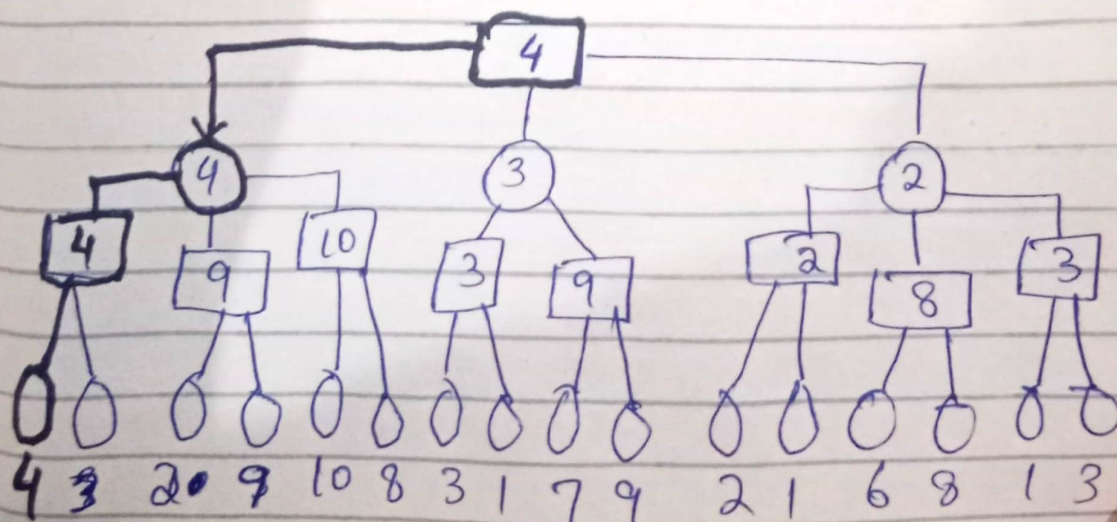
Step (3) Max-turn: level (1)

⇒ max has 1 option out of 3 possible moves.

max will choose $\max(4, 3, 2)$
so 4 it is.

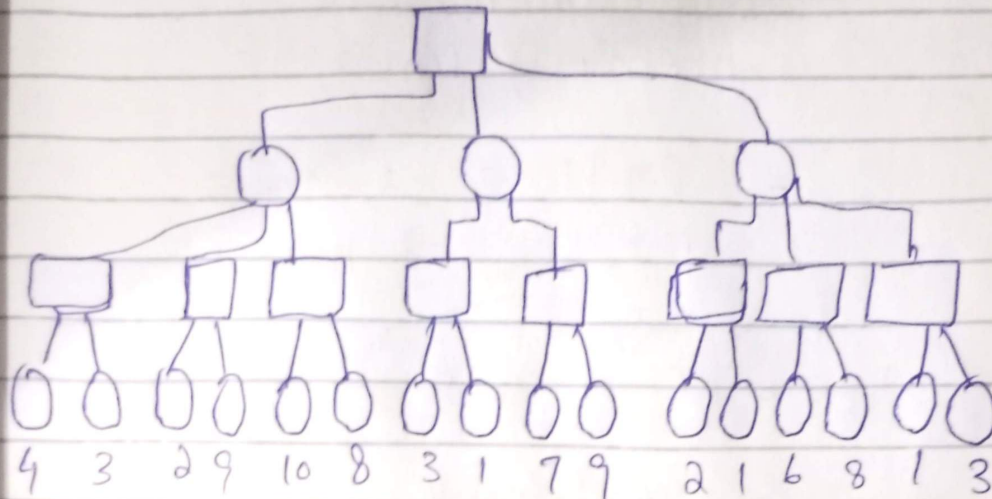


final dept-first-search tree.



QUESTION : 2 Alpha Beta:

initial tree:

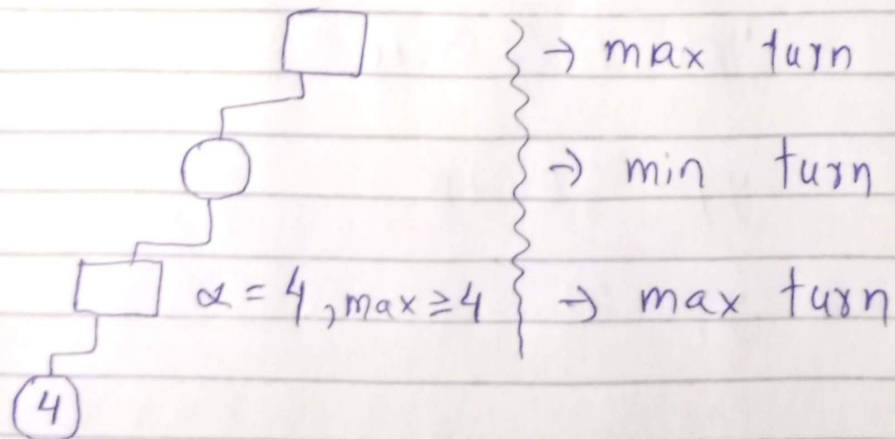


Step = 1:

DFS

L.H.S:

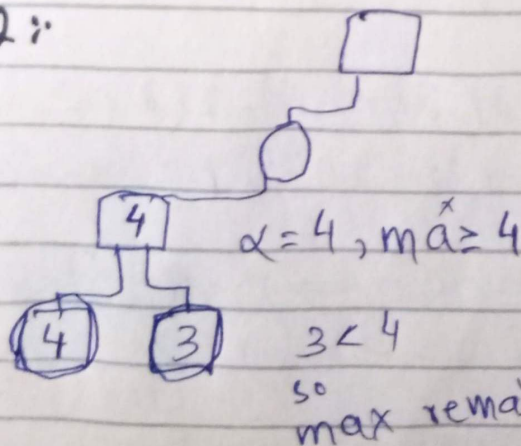
max, turn



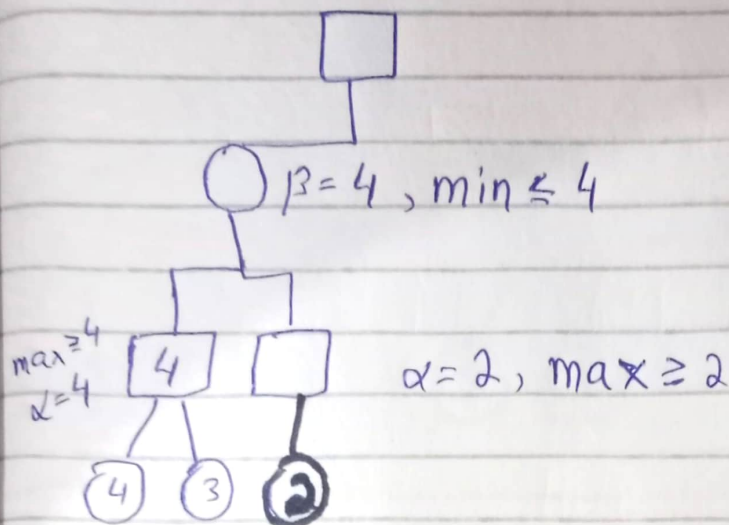
Step : 2:

{ max
turn }

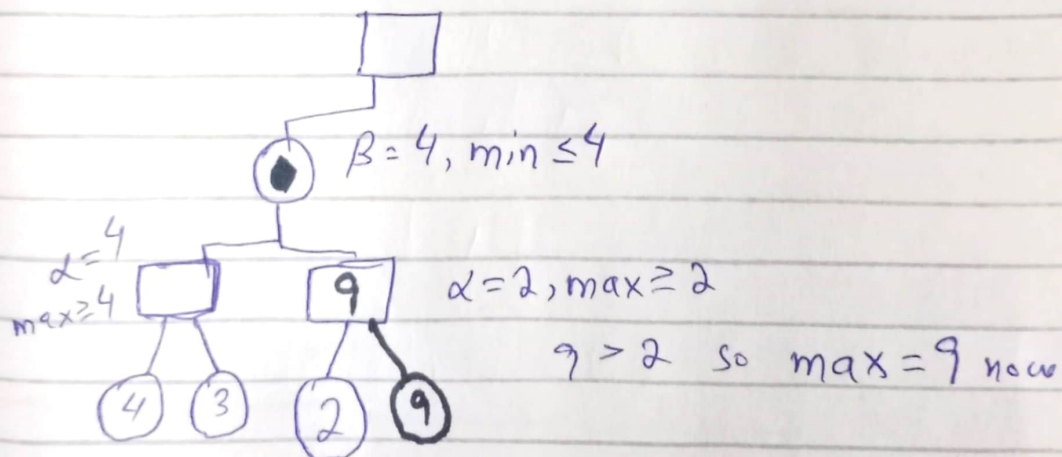
exploring
2nd leaf



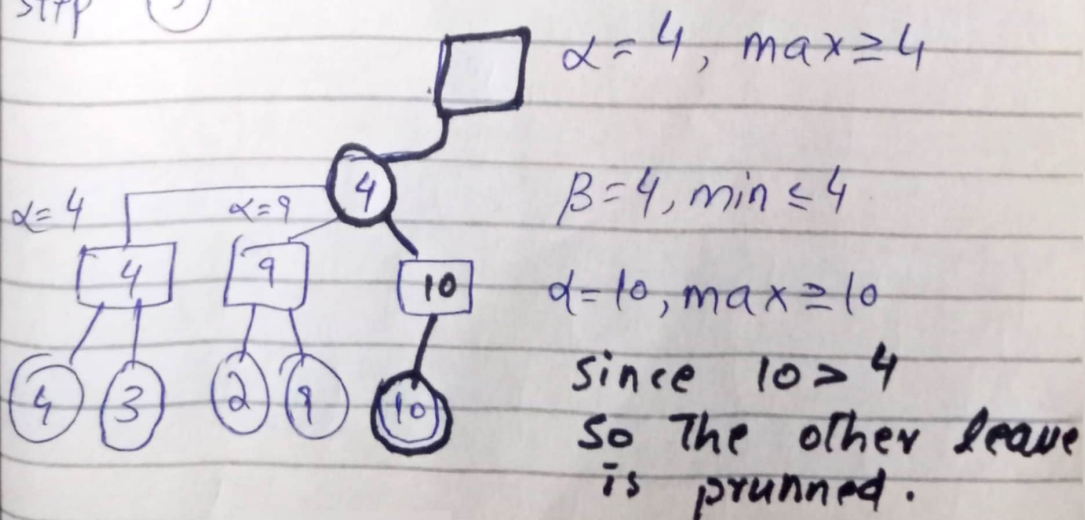
Step 3:



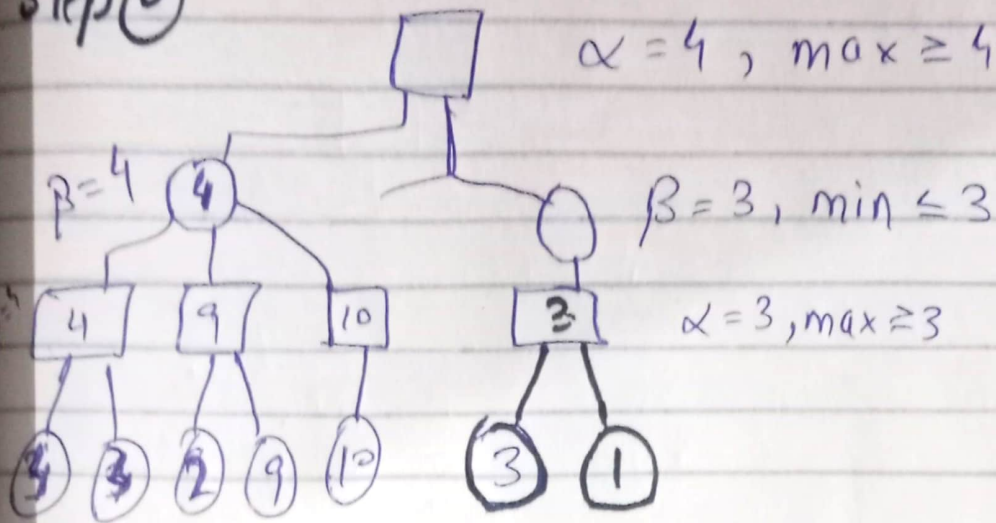
Step 4:



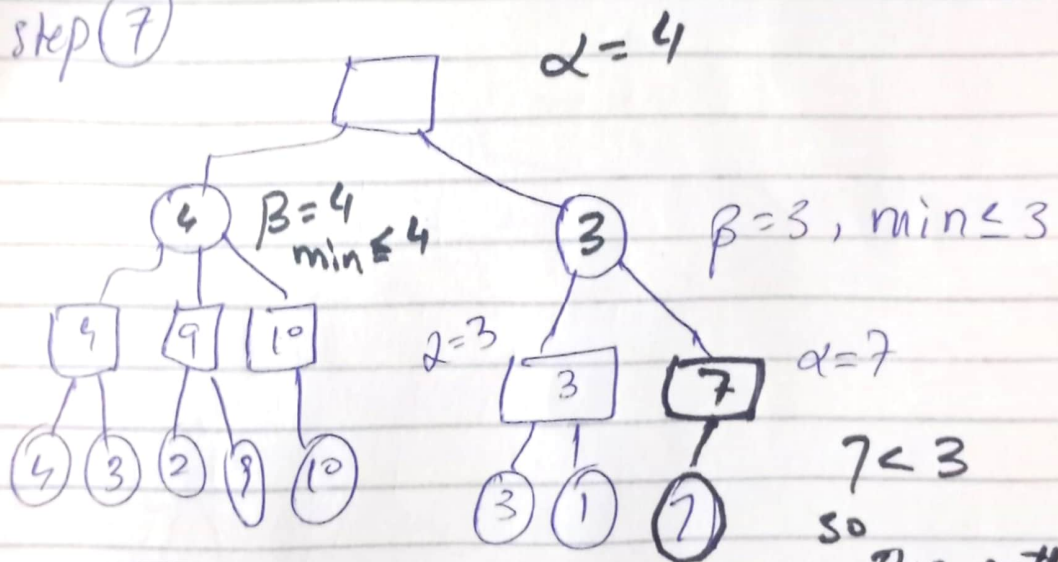
Step 5



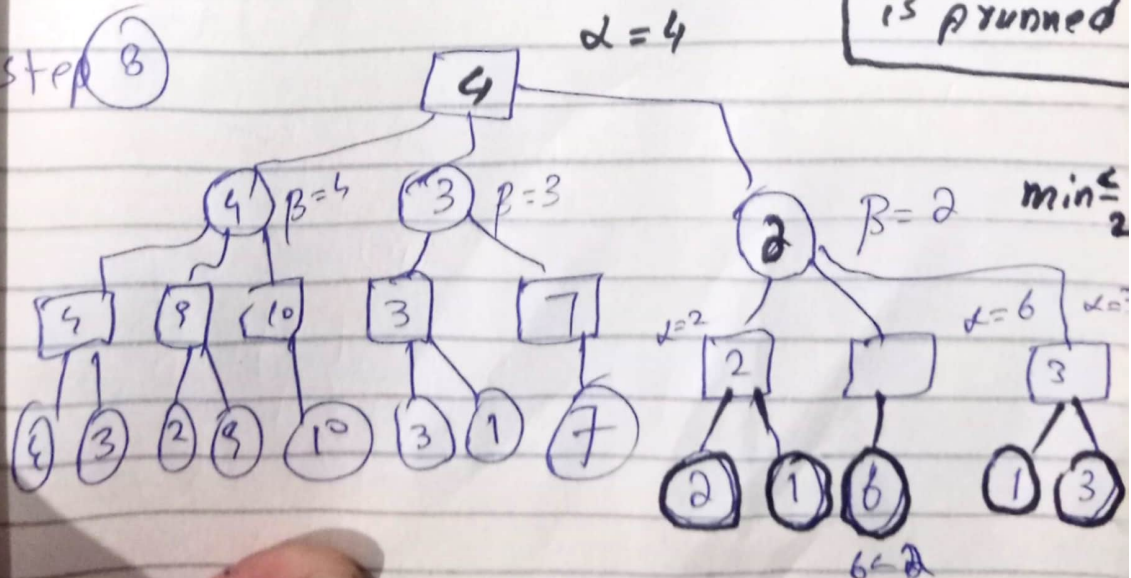
Step (6)



Step (7)



Step (8)



~~when~~ Last step:

So the final value of
root node = 4.

and dfs tree is the
left most branch of the
tree.

